

in that sulphur-rich, superheated hostile environment... There's reason to believe that life exists in extreme conditions, and there's evidence to prove that Mars has ice, water vapour and, albeit for short stints, liquid water. There are also huge caverns on Mars, so deep that natural light from the sun never reaches their bottoms, and these are well-shielded from the extreme climate that ravages Mars' surface. The bottom-line, however, is that we'll never know—not until we get there ourselves and have a look-see.

Problems

Advances in technology are shaping our future, making the impossible possible. People with no legs are running races, dead hearts are being replaced by machine pumps, people are video-conferencing from across the globe, bombs more destructive than fission reactions and that leave no long-term radiation effects have been tested... Trillions of dollars are being spent on defence—that's a 13-figure number—by various countries, and over a billion people have been exposed to PCs. It's all happening, and happening now.

Those of us old enough to read this magazine have seen more change in our lifetimes than a human of any other time has ever seen since the beginning of recorded history.

That digression was necessary to try and give you a sense of wonderment at our achievements, some pride in the capabilities of the human brain... mainly so that we could bring you crashing down again, to reveal to you a harsh reality: man went to the moon decades ago, and since then has been content to stay right here. Impetus in space explorations has slowed to a crawl when compared to the space race of the sixties and seventies. We're stirring though, looking upwards, dreaming of distant worlds—again.

The maximum distance from the Earth to the Moon is roughly a little over 400,000 km; the *minimum* distance from Earth to Mars, however, occurred on 27 August 2003, after 60,000 years, when Earth and Mars were a mere 56,000,000 km apart. Think of those numbers as 4 *lakh* and 5.6 *crore*, and you will better appreciate the vast distances involved in landing a human on Mars.

Complexities

It's all well and good to sit here in our comfortable chairs and theorise, day-dream and wonder, but the task of attempting to put a man on Mars is several magnitudes more complex than the famed Moon landing. To put it simply, aiming a rocket at the moon seems easy when you start thinking of travelling to Mars. There are problems, and we're looking to technology to solve them...

Navigation in space is quite different and much, much more complex than most of us imagine. There's gravitational pulls from planets and the sun, meteors, comets, space dust, space junk, and more to avoid. Add to that the fact that you're pointing this tiny speck of a craft—launching it from one planet, moving it at break-neck speeds—at another planet that's hurtling through space, and you get complexities and probabilities that will leave even the best mathematicians amongst us scratching their heads.

Propulsion

When you're sending machines, it's a lot easier, because time isn't of the essence. Robots and Rovers don't get bored, they don't need to be fed, and can be turned off to save power. Most importantly, they can just be left there. A human, on the other hand, has to get there quickly, safely, and then has to be brought back.

It's obvious that NASA is leading the world in space exploration technologies, and it has already sent various probes to Mars—even landing the Mars Rovers, Spirit and Opportunity, on the Martian surface in 2003. As of now, the two Rovers have overcome all odds and are still exploring the Martian surface, over three years after their intended 90-day life expectancy passed.

It takes quite a big rocket to launch even a small rover into space—consider that the rockets that launched the Rovers weighed about 2,85,000 kg, of which a mere 1,000 kg was the actual spacecraft that went on to Mars. The fuel and oxygen are used to create the burn at lift off, and a complicated turbo-turbine pump ensures that the amount of fuel and oxygen increases in the engine to break away from the Earth's gravity. The rocket also contains complicated gimbals and gyroscopes to make sure that the rocket stays on its path upwards. (Gimbals are devices used to ensure that an object is always kept horizontal.) The entire rocket engine is mounted on gimbals, and is thus actually movable.

Now there are two smaller engines on either side of the rocket that fire only when the rocket is going off course or tilting towards one side. These also work with the gimbals to ensure that the rocket stays pointed straight up during the launch. As if that weren't enough, there are the nine additional rockets attached to the side of the rocket to provide additional thrust—think nitro boosters of popular car racing games. These nine rockets run on a solid fuel called hydroxyl-terminated polybutadiene (HTPB), held in shells made of graphite-epoxy—up to five times lighter than most metals. That completes Stage I of the rocket motors.

When the rocket is about 130 km above Earth, Stage II kicks in. Stage II is made of a rocket that burns a mixture of Aerozine 50 (50% hydrazine, 50% unsymmetric dimethyl-hydrazine) and nitrogen tetroxide or N₂O₄ (an oxidiser). Because of the lack of an atmosphere in space, fuels that need ignition cannot be used, and this second-stage fuel is made just for that—the fuel automatically burns when the ingredients are mixed. The Stage II rocket fires once in order to get the spacecraft into orbit around Earth. Only after all the calculations are done, and the Stage II rocket is lined up with its final destination, does it fire again. Here, too, gimbals and gyroscopes are used to make sure that the spacecraft stays on course. Computers onboard the spacecraft do all the manoeuvring and rocket firing.

Power Generation

Many of us Indians are quite used to power cuts, and life just goes on. However, millions of kilometres away from Earth, with no free air to breathe, a spacecraft to man, running computers that an astronaut's life depends on, power short-

